

Post hoc bounds for heterogeneous tests

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- **Post hoc bounds** : Allows users to choose the rejection set freely : a way to handle p -hacking.
- **Heterogeneous data** : Consider the natural heterogeneity of data to gain power in post hoc procedures. This heterogeneity arise naturally in discrete data or when performing weighting.

Post hoc bounds

Let $X \in \mathcal{X}$ be a sample of law $\mu \in \mathcal{P}$ where $\mathcal{P}$ is a set of probability measures.

Let $H_1, \dots, H_m \subset \mathcal{P}$ be m null hypotheses.

Each hypothesis i has an associated p -value, $p_i(X)$ such that, if $\mu \in H_i$,

$F_i(\alpha) := \mathbb{P}_{X \sim \mu}(p_i \leq \alpha) \leq \alpha$ for all $\alpha \in [0, 1]$, and we note

$\mathcal{H}_0(\mu) = \{i : \mu \in H_i\}$.

Definition 1 (Post hoc bounds)

Let $\alpha \in (0, 1)$. A post hoc bound of level α is a statistic $\hat{V}_\alpha : \mathcal{P}(\mathbb{N}_m) \rightarrow \mathbb{N}$ such that, for all $\mu \in \mathcal{P}$,

$$\mathbb{P}_{X \sim \mu}(\forall R \subset \mathbb{N}_m, V(R) \leq \hat{V}_\alpha(R)) \geq 1 - \alpha,$$

where $V(R) = |\mathcal{H}_0(\mu) \cap R|$.

Definition 2 (JER)

For a reference family $\mathcal{R} = ((R_k, \zeta_k))_{1 \leq k \leq K}$ and $\mu \in \mathcal{P}$, the joint error rate control is :

$$JER(\mathcal{R}, \mu) := \mathbb{P}_{X \sim \mu}(\exists k \in \mathbb{N}_K : V(R_k) > \zeta_k).$$

Allows to compute a valid post hoc bound with interpolation when the JER is upper bounded by α (Blanchard et al., 2020, Durand et al., 2020).

Proposition 1 (Interpolated bound, Blanchard et al., 2020)

Let $\mathcal{R} = ((R_k, \zeta_k))_{1 \leq k \leq K}$ a family with JER bounded by $\alpha \in [0, 1]$ and $\mathcal{A}(\mathcal{R}) := \{A \subset \mathbb{N}_m : \forall k \in \mathbb{N}_K, |R_k \cap A| \leq \zeta_k\}$.
Then $V_{\mathcal{R}}^*(R) := \max_{A \in \mathcal{A}(\mathcal{R})} |R \cap A|$ is a level α post hoc bound.

The following settings are classical *JER* settings :

- **Top- k setting** : $R_k = \{\sigma(i), 1 \leq i \leq k\}$ (where σ orders the p_i),
- **k -FWER setting** : $\zeta_k = k - 1$,
- **Deterministic setting** : the R_k are deterministic and have a forest structure, meaning that for all $(k, k') \in \{1, \dots, K\}^2$,
 $R_k \cap R_{k'} \in \{\emptyset, R_k, R_{k'}\}$.

Let $\alpha \in [0, 1]$ and, for $A \subset \{1, \dots, m\}$, $H_A := \bigcap_{i \in A} H_i$ the intersection hypothesis for A .

Suppose to have at hand, for all $A \subset \{1, \dots, m\}$, a local test φ_A , and denote $\mathcal{N}(X) = \{A \subset \mathbb{N}_m : \varphi_A(X) = 0\}$.

Proposition 2 (Inversion procedure, Genovese and Wasserman, 2006)

Suppose that $X \sim \mu$ and $\varphi_{\mathcal{H}_0(\mu)}$ is a test of level α . Then, the statistic $\hat{V}_{\mathcal{N}}$ defined by $\hat{V}_{\mathcal{N}}(R) := \max_{A \in \mathcal{N}} |A \cap R|$ is a post hoc bound of level α .

Local tests from concentration inequalities

Suppose that for all $A \subset \{1, \dots, m\}$, $\varphi_A = \sup_{t \in [0,1]} \varphi_{A,t}$.

Lemma 1

For all $R \subset \mathbb{N}_m$,

$$\hat{V}_{\mathcal{N}}(R) = \max \left\{ \ell \in \{0, \dots, |R|\} : \min_{A: |A \cap R| = \ell} \sup_{t \in [0,1]} \varphi_{A,t} = 0 \right\}.$$

Lemma 2

Denote by $\varphi_{R,\ell}^{SC1} := \max_{t \in [0,1]} \min_{A: |A \cap R| = \ell} \varphi_{A,t}$, then :

$$\begin{aligned} \hat{V}_{\mathcal{N}}(R) &\leq \max \left\{ \ell \in \{0, \dots, |R|\} : \varphi_{R,\ell}^{SC1} = 0 \right\} \\ &\leq \min_{t \in [0,1]} \left\{ \max \left\{ \ell \in \{0, \dots, |R|\} : \min_{A: |A \cap R| = \ell} \varphi_{A,t} = 0 \right\} \right\}. \end{aligned}$$

Concentration inequalities

Homogeneous inequalities (1/2)

Proposition 3 (DKWM inequality - Improved version from Reeve, 2024)

Let $X_1, \dots, X_m \stackrel{iid}{\sim} F$ and $\hat{F}_m(t) := \frac{1}{m} \sum_{1 \leq i \leq m} \mathbb{1}_{X_i \leq t}$ for all $t \in \mathbb{R}$.

Then, for all $\lambda \geq 0$,

$$\mathbb{P}\left(\sqrt{n} \sup_{t \in \mathbb{R}} (\hat{F}_m - F)(t) > \lambda\right) \leq \exp(-2\lambda^2),$$

leading, when applied to $(p_i)_{i \in A} \stackrel{iid}{\sim} \mathcal{U}([0, 1])$, to $\varphi_{A,t} = \mathbb{1}_{\{\sqrt{|A|}(\hat{F}_A(t) - t) > \lambda\}}$.

Homogeneous inequalities (2/2)

Proposition 4 (Simes inequality)

Suppose $(p_1, \dots, p_m)$ verifies the wPRDS property under $\mu \in \bigcap_{1 \leq i \leq m} H_i$. Then for all $\alpha \in [0, 1]$, $\mathbb{P}_{X \sim \mu}(\exists k \in \{1, \dots, m\} : p_{(k)} \leq \frac{\alpha k}{m}) \leq \alpha$.

Proposition 5 (Uniform version of Wellner inequality, Meah et al., 2024)

Let $U_1, \dots, U_m \stackrel{iid}{\sim} \mathcal{U}([0, 1])$ and $\kappa = \frac{\pi^2}{6}$. For all $\delta \in (0, 1]$,
$$\mathbb{P}\left(\forall t \in [0, 1], m^{-1} \sum_{1 \leq i \leq m} \mathbb{1}_{U_i \leq t} \leq t h^{-1}\left(\frac{2 \log(\kappa/\delta) + 4 \log(1 + \log_2(1/t))}{mt}\right)\right) \geq 1 - \delta,$$

with $h(\lambda) = \lambda(\log \lambda - 1) + 1$.

Proposition 6 (Bretagnolle's inequality, Shorack and Wellner, 1986 with the improvement from Reeve, 2024)

Let $X_1, \dots, X_m$ be independent, denote by F_i the cdf of X_i and $\bar{F}_m$ the mean of F_i 's.

Then, for all $\lambda \geq 0$,

$$\mathbb{P}\left(\sqrt{m} \sup_{t \in \mathbb{R}} (\hat{F}_m - \bar{F})(t) > \lambda\right) \leq \exp(1 - 2\lambda^2).$$

Proposition 7 (Simes-type inequality derived from Döhler et al., 2018)

Suppose that, for $\mu \in \bigcap_{1 \leq i \leq m} H_i$, $p_1, \dots, p_m$ is a family of m independent p -values with known cdf $F_1, \dots, F_m$.

Let $\mathbf{H}(s, t) = \left(\frac{F_i(t)}{1 - F_i(s)} \right)_{1 \leq i \leq m}$ for all $(s, t) \in [0, 1[\times [0, 1]$, $\alpha \in]0, 1[$ and define

$\tau_m := \max \left\{ t \in \mathcal{A} : \sum_{1 \leq k \leq m} \mathbf{H}(t, t) \leq m\alpha \right\}$, and, for all $k \in \{1, \dots, m-1\}$,

$\tau_k := \max \left\{ t \in \mathcal{A}, t \leq \tau_m : (\mathbf{H}(\tau_m, t))_{[1]} + \dots + (\mathbf{H}(\tau_m, t))_{[m-k+1]} \leq k\alpha \right\}^1$.

Then, $\mathbb{P}_{X \sim \mu}(\exists k \in \{1, \dots, m\} : p_{(k)} \leq \tau_k) \leq \alpha$.

¹ $x_{[i]}$ is the i -iest larger coordinate from x .

Heterogeneous-adapted post hoc bounds and numerical experiments

Application to top- k setting

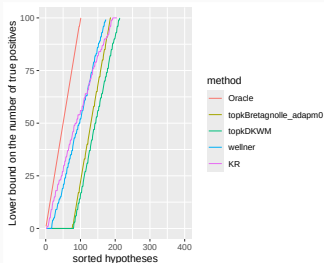
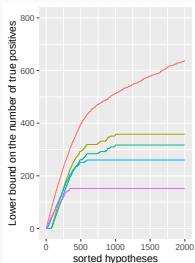
Theorem 1 (Bretagnolle's top- k calibration)

Let $\alpha \in (0, 1]$. Suppose that $(p_i)_{i \in \mathcal{H}_0}$ is an independent family of p -values.

Define $\lambda = \sqrt{\frac{1 + \log(1/\alpha)}{2}}$ and

$$\zeta_k = k \wedge \left\lfloor m \bar{F}(p_{(k)}) + \sqrt{m} \lambda \right\rfloor.$$

Then, the reference family $\mathfrak{R} = ((R_k, \zeta_k))_{k \in \mathcal{K}}$ has its JER bounded by α .



Application to deterministic setting

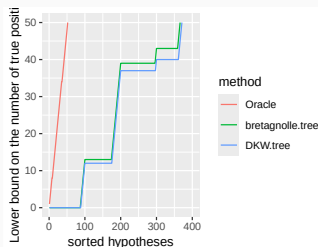
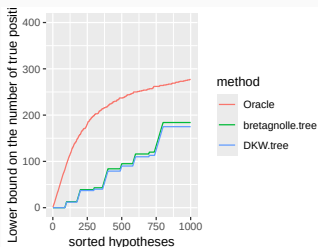
Theorem 2 (Bretagnolle's estimation of m_0)

Let $(R_k)_{k \in \mathcal{K}}$ be a deterministic family and $\alpha \in]0, 1[$. Suppose that for all $k \in \mathcal{K}$, $(p_i)_{i \in \mathcal{H}_0 \cap R_k}$ is an independent family of p -values. Define

$\lambda = \sqrt{\frac{1 + \log(K/\alpha)}{2}}$, $\mathbb{G}$ such that $\mathbb{G}(t) = (\mathbb{1}_{p_i \leq t} - F_i(t))_{1 \leq i \leq m}$ for all $t \in [0, 1]$, and

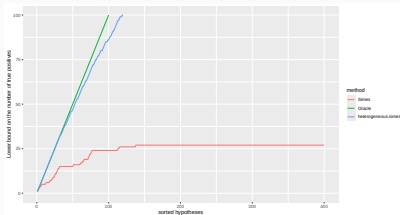
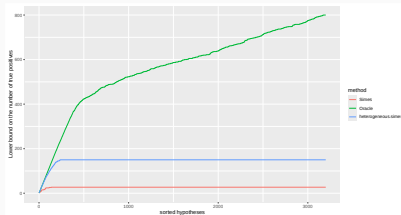
$$b_k(p_i) = \max \left\{ \ell : \ell^{-1/2} \left(\sum_{1 \leq j \leq \ell} (\mathbb{G}(p_i))_{(j)} \right) \leq \lambda \right\}.$$

Then, the reference family $\mathfrak{R} = ((R_k, \zeta_k))_{k \in \mathcal{K}}$, with $\zeta_k = \min_{i \in R_k} b_k(p_i)$, has its JER bounded by α .



Theorem 3 (Simes-type calibration)

Let, for all $k \in \mathbb{N}_m$, $\zeta_k = k - 1$ and $R_k = \{i \in \mathbb{N}_m : p_i \leq \tau_k\}$. Then the reference family $((R_k, \zeta_k))$ has its JER bounded by α .



- Have a broader set of simulations and apply with real data sets,
- Get a suitable heterogeneous version of Wellner inequality,
- Get a heterogeneous version of Katsevich and Ramdas inequality.

-  Blanchard, Gilles, Pierre Neuvial, and Etienne Roquain (2020). “Post hoc confidence bounds on false positives using reference families”. In: *Ann. Statist.* 48.3, pp. 1281–1303. ISSN: 0090-5364, 2168-8966. DOI: 10.1214/19-AOS1847. URL: <https://doi.org/10.1214/19-AOS1847>.
-  Döhler, Sebastian, Guillermo Durand, and Etienne Roquain (2018). “New FDR bounds for discrete and heterogeneous tests”. In: *Electron. J. Stat.* 12.1, pp. 1867–1900. ISSN: 1935-7524. DOI: 10.1214/18-EJS1441. URL: <https://doi.org/10.1214/18-EJS1441>.
-  Durand, Guillermo, Gilles Blanchard, Pierre Neuvial, and Etienne Roquain (2020). “Post hoc false positive control for structured hypotheses”. In: *Scand. J. Stat.* 47.4, pp. 1114–1148. ISSN: 0303-6898, 1467-9469. DOI: 10.1111/sjos.12453. URL: <https://doi.org/10.1111/sjos.12453>.
-  Durand, Guillermo, Florian Junge, Sebastian Döhler, and Etienne Roquain (2019). *DiscreteFDR: An R package for controlling the false discovery rate for discrete test statistics*. arXiv: 1904.02054 [stat.CO]. URL: <https://arxiv.org/abs/1904.02054>.

-  Genovese, Christopher R. and Larry Wasserman (2006). “Exceedance control of the false discovery proportion”. In: *J. Amer. Statist. Assoc.* 101.476, pp. 1408–1417. ISSN: 0162-1459,1537-274X. DOI: 10.1198/016214506000000339. URL: <https://doi.org/10.1198/016214506000000339>.
-  Katsevich, Eugene and Aaditya Ramdas (2020). “Simultaneous high-probability bounds on the false discovery proportion in structured, regression and online settings”. In: *Ann. Statist.* 48.6, pp. 3465–3487. ISSN: 0090-5364,2168-8966. DOI: 10.1214/19-AOS1938. URL: <https://doi.org/10.1214/19-AOS1938>.
-  Meah, Iqraa, Gilles Blanchard, and Etienne Roquain (2024). “False discovery proportion envelopes with m -consistency”. In: *J. Mach. Learn. Res.* 25, Paper No. 270, 52. ISSN: 1532-4435,1533-7928.
-  Reeve, Henry W J (2024). *A short proof of the Dvoretzky–Kiefer–Wolfowitz–Massart inequality*. arXiv: 2403.16651 [math.PR].



Shorack, Galen R. and Jon A. Wellner (1986). *Empirical processes with applications to statistics*. Wiley Series in Probability and Mathematical Statistics: Probability and Mathematical Statistics. John Wiley & Sons, Inc., New York, pp. xxxviii+938. ISBN: 0-471-86725-X.

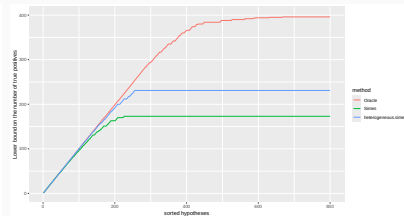
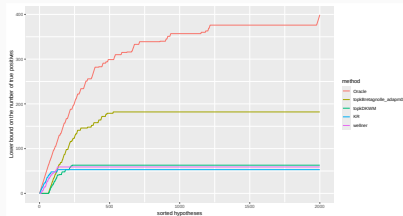
Theorem 4

Let $\alpha \leq 0.31$. Then, in the top- k setting with p -values under the null independent,

$$\zeta_k := \left\lfloor \frac{\log(1/\alpha)}{\log(1 + \log(1/\alpha))} (1 + mp_{(k)}) \right\rfloor$$

gives a reference family with JER bounded by α .

Simulations with Fisher exact test



Artificial setting:

- **top- k** : $m = 3200$ and $m_0 = 2400$. Continuous p -values sampled from a Gaussian model, with mean equal to 0 under H_0 and 3 under H_1 . Each p -value is discretized using a support of length 50 sampled from a uniform law.
- **k-FWER** : Same simulation setting with the support equal to 300.
- **Deterministic setting** : $m = 3200$ and $m_0 = 2800$, 8 active leaves of length 100 and each active leaf contains 50% of p -values under H_1 .

Fisher exact test:

Same simulation settings than Döhler et al., 2018, with $N = 50$, $m = 2000$, $\pi_0 = 0.8$ and the response under H_1 simulated by binomials of parameter 0,25 (0,4 for Simes' simulations) against 0.1 under H_0 .